WORKSHEET

Chapter 8: Implicit differentiation

1 Find the derivative of each of the following.

a $y^2 + x - y^3 + 6 = 3y$

b $y^3 + y = \sqrt{x^2 + y^2}$

c $xy + 2x + 3x^2 = 4$

d $x^3 + y^3 = x^2 + y^2$

e $x^3 + y^3 = 6xy$

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f $\sin(x) + \cos(y) = -2y^3$

2 Consider the function $xy = 1$.

a Differentiate the function implicitly to find $\frac{dy}{dx}$.

b Find y explicitly in terms of x and show that $\frac{dy}{dx}$ is the same as that in part **a**.

3 Find the derivative of each of the following.

a $5y^2 = 2x^3 - 5y$

b $x^2 + y = 1 + y^3$

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c $1 = 3x + 2x^2y^2$

d $2y = x^2 + \sin(y)$

e $3x^2y^2 = 4x^2 - 4xy$

f $x^2 - xy - y^2 - 2y = 0$

4 Find $\frac{dy}{dx}$ in terms of x and y for each of the following by implicit differentiation.

a $\cos(y) = x + y$

b $e^y = xy$

c $\sin(y) + x^2 + 4y = \cos(x)$

d $e^x + y = e^{xy} + 2$

e $\sin(y) + x^2y^3 - \cos(x) = 2y$

f $y^2 + x^3 - xy + \cos(y) = 0$

5 Find the equations of the tangents to the parabola with equation $x = y^2$ at the points $(4, 2)$ and $(4, -2)$.

6 Find the equation of the normal to the curve with equation $x^2y^2 + y = 2$ at the point $(1, -2)$.

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7 Find the equation of the tangent to the curve with equation $2x^2 + 3y^2 = 5$ at (1, 1).

8 Find the equation of the tangents to the curve with equation $xy^2 - 5x = 2(y^2 + x^2y - 16)$ where $x = 3$.

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Answers

1 a $\dfrac{3x^2}{3-2y+3y^2}$

b $\dfrac{x}{(y+y^3)(3y^2+1)-y}$

Alternatively:

$$= \frac{x}{(3y^2+1)(x^2+y^2)^{\frac{1}{2}}-y}$$

$$= \frac{x}{3y^5 4y^3}$$

$$= \frac{\sqrt{y^2+2}}{3y^3+4y}$$

c $\dfrac{-2-6x-y}{x}$

d $\dfrac{-3x^2+2x}{3y^2-2y}$

e $\dfrac{2y-x^2}{y^2-2x}$

f $\dfrac{\cos(x)}{\sin(y)-6y^2}$

2 a $\dfrac{dy}{dx} = \dfrac{-1}{x^2}$

b $y = x^{-1}$, $\dfrac{dy}{dx} = -x^{-2} = \dfrac{-1}{x^2}$

3 a $\dfrac{6x^2}{10y+5}$

b $\dfrac{2x}{3y^2-1}$

c $\dfrac{-3-4xy^2}{4x^2y}$

d $\dfrac{2x}{2-\cos(y)}$

e $\dfrac{4x-2y-3xy^2}{3x^2y+2x}$

f $\dfrac{2x-y}{x+2y+2}$

4 a $\dfrac{dy}{dx} = \dfrac{1}{-\sin(y)-1}$

b $\dfrac{dy}{dx} = \dfrac{y}{e^y-x}$

c $\dfrac{dy}{dx} = \dfrac{-\sin(x)-2x}{4+\cos(y)}$

d $\dfrac{dy}{dx} = \dfrac{ye^{xy}-e^x}{1-xe^{xy}}$

e $\dfrac{dy}{dx} = \dfrac{2xy^3+\sin(x)}{2-\cos(y)-3x^2y^2}$

f $\dfrac{dy}{dx} = \dfrac{y-3x^2}{2y-\sin(y)-x}$

5 $y = \frac{1}{4}x + 1$ at (4, 2) and

$y = \frac{-1}{4}x - 1$ at (4, −2).

6 $y = \frac{-3}{8}x - \frac{13}{8}$

7 $y = \frac{-2}{3}x + \frac{5}{3}$

8 $y = -5x + 32$ and $y = -x + 2$

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